Project Design Phase
Solution Architecture

Date	08 October 2026
Team ID	SWTID-2026-2540
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Solution Architecture:

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Script-Controlled ACL – Restrict Record Access Based on Field Value

Architecture Overview

The proposed solution is implemented entirely within ServiceNow using a custom Institution Details table, user roles, record-level Access Control Lists (ACLs), a Branch field condition, and an advanced script for READ authorization. The architecture separates authentication/role identification from authorization of each CRUD operation. The READ layer evaluates the EEE branch requirement, while CREATE, WRITE and DELETE are controlled by bb2, bb3 and bb4 respectively. Administrators retain unrestricted access.

Solution Architecture and Access Flow

Layer / Component	Implementation	Access Decision / Flow
1. User	EEE User / Administrator	User signs in to ServiceNow; roles are evaluated.
2. Roles	bb1, bb2, bb3, bb4	bb1 → read; bb2 → create; bb3 → write; bb4 → delete.
3. Data Source	u_institution_details	Stores student/institution records and the Branch value.
4. Branch Data	ECE / EEE / CSE	READ access is restricted to records where Branch = EEE.
5. READ ACL	Record ACL + Data condition + Advanced script	Admin → allow. bb1 → allow for EEE records. Others → deny.
6. CREATE ACL	Record ACL requiring bb2	Only users with bb2 can create records.
7. WRITE ACL	Record ACL requiring bb3	Only users with bb3 can modify existing records.
8. DELETE ACL	Record ACL requiring bb4	Only users with bb4 can delete records.
9. Validation	Impersonation + list/form testing	Verify EEE-only visibility and CRUD permissions for each role.

ACL Configuration

Operation	ACL Name	Required Role	Condition / Script	Expected Result
READ	u_institution_details	bb1	Branch = EEE + advanced script; admin is allowed.	bb1 users see EEE records; admin sees all.
CREATE	u_institution_details	bb2	No additional data condition.	bb2 users can create records.
WRITE	u_institution_details	bb3	No additional data condition.	bb3 users can edit records they are authorized to access.
DELETE	u_institution_details	bb4	No additional data condition.	bb4 users can delete records they are authorized to access.

READ ACL Script Logic

```
(function () {
    if (gs.hasRole('admin')) {
        return true;
    }
    if (gs.hasRole('bb1')) {
        return true;
    }
    return false;
```

```
})();
```

The Branch = EEE data condition limits the READ ACL to EEE records. The script grants the administrator full access and permits the bb1 role; users who do not satisfy the ACL requirements are denied.

End-to-End Flow

1. User logs into ServiceNow.
2. ServiceNow identifies the user's assigned roles.
3. User requests the u_institution_details list or a record operation.
4. ServiceNow evaluates the applicable ACL for READ, CREATE, WRITE or DELETE.
5. For READ, the Branch condition and advanced script are evaluated.
6. If the ACL permits the operation, the requested record/action is available; otherwise access is denied.
7. Impersonation testing verifies EEE-user and administrator behavior.

Key Components and Requirements

Component	Purpose
ServiceNow Platform	Hosts the table, users, roles, ACLs, forms and lists.
Custom Table	u_institution_details with Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number and Description.
Roles	bb1, bb2, bb3 and bb4 provide operation-specific authorization.
ACL Engine	Evaluates record-level permissions before allowing the requested operation.
Validation	Use impersonation to verify EEE-only read access and CRUD permissions.

